Assignment 3: Unsupervised Learning - Report

IAT461 - Summer 2026 - Stin (S.D)

A1

1. **Records with No Geographic Coordinates: `geom` `geo_point_2d` `geometry` & `latitude` `longitude`

We can't apply K-Means or DBScan on locations without coordinations, therefore these should be removed

2. **Combine businesstype and businesssubtype into a single, more fine-grained industry label where a subtype exists

First I'll standardize the typecase for all the entries + clean up bad entries

Then I'll combine the subtype with a '-' if it exists. Otherwise, do nothing.

3. **Group by `business_category` and then input missing `numberofemployees` with that domain-specific median.

This prevents heavy right-skewing since the businesses themselves pull the mean upwards when they record their number of employees. The median ensures missing values are realistically standardized to be resilient to the outliers

5. **Do the same to the `feepaid`

4. **Filter to a sensible subset (e.g., status == "Issued") and justify

Since `Gone Out of Business`, `Inactive`, & `Cancelled` can be considered noise and phantom clusters, we filter by issued so every datapoint is valid.

`pending` is removed because the application process was never finalized and thus usually have `feepaid` as 0. Removing them prevents distorting feature scaling.

A2 — Location-only clustering: K-means vs. DBSCAN

How do the K-means clusters map onto actual Vancouver geography — do they correspond to anything you'd recognize (downtown core, west side, etc.)?

For K-Means = 4, the city is divided into four regions around its centre. The dark blue cluster looks like the Grandville and Downtown area, while the pink one looks like Kingsway or East Van.

****How do the DBSCAN clusters compare? Where do the two methods agree or disagree, and why — think about what each algorithm assumes about cluster shape and density, and what DBSCAN's "noise" points represent.****

While K-Means forces 4 equal parts of the city to be segmented, DBSCAN gives us a more accurate idea of the actual density of commercial Vancouver businesses without assuming anything.

They agree that Downtown Vancouver is the primary most dense business hub of all the other groups.

The main disagreement is on what shapes separate groups make up. K-Means takes into account all dots but DBSCAN prioritizes finding clusters while disregarding other points as noise.

A3 — Feature-based clustering: Size, Industry, Lifecycle

****What do the resulting clusters seem to represent? Does this differ from the purely geographic clustering in A2?****

Cluster 3 (Cyan) - Negative PCA values, where PCA < -3.0
- Large retail stores with lots of employees, hotels, and other types of high-fee chains

Cluster 0 (Blue) - Central, -3 < PCA < -0.5
- Mid-scale businesses and standard stores. Restaurants with standard employee numbers and fees.

Cluster 1 (Red) - Positive PCA values, 0 < PCA < 1.0
- Small business, single person services & small trades businesses

Cluster 2 (Pink) - High Positive PCA values, > 1.0
- Extremely specific one-hot encoded stores with unpredictable license insurance patterning

This feature-based clustering differs from the geographic clustering (K-means, DBscan in A2) because its on operational scale and power instead of physical location. So even if two banks aren't anywhere near each other but have the same value of features, they're clustered together.

PCA also tells us the variances in features. Where X = Business scale/size, Y = Lifecycle

B4

****Does the grouping seem meaningful given what you know (or can look up) about these areas? Any surprising groupings? Explain using the actual cluster membership, not just the general shape of the plot.****

Neighborhoods with similar business environments naturally pair up. For example, commercial and office hubs like **Downtown** and **Fairview** group together because they are full of corporate offices, shops, and restaurants. They are touristy destinations.

Quieter residential areas like **Dunbar-Southlands** and **Kerrisdale** form their own group, made up mostly of local health clinics, personal services, and small home businesses.